BRAM MODESTUS NAIBAHO



Full Stack Developer | IoT & Cloud Computing

Tuktuk Siadong, North Sumatra, Indonesia • +62 823-7369-8610 • brammodestusnaibaho@gmail.com
bram-modestus-naibaho.vercel.app • github.com/Bram-02
linkedin.com/in/bram-modestus-naibaho-95866b387

PROFILE

Diploma III Computer Technology student at Institut Teknologi Del who builds end-to-end systems: web applications (Laravel, React, Node.js, MySQL) and LoRa-based IoT monitoring (ESP32, Raspberry Pi, MQTT, InfluxDB, Grafana). Led a three-person team as Project Manager & System Analyst on a solar-powered Lake Toba water quality buoy, and is now extending it into a second-generation, cloud-connected version. Also developing a village information system (community service) and a restaurant website with online ordering. Solid foundation in networking and cybersecurity (Huawei HCIA-Datacom, Cisco Ethical Hacker). Seeking an internship or entry-level role in full-stack web development, IoT, or cloud computing.

TECHNICAL SKILLS

Languages	Python, JavaScript, PHP, Java, C++
Frontend	HTML, CSS, React
Backend	Node.js, Laravel, REST APIs, MQTT
Databases	MySQL, PostgreSQL, MongoDB, InfluxDB (time-series)
IoT & Embedded	ESP32, LoRa, Raspberry Pi, Arduino IDE; sensors: pH, temperature (DS18B20), TDS, turbidity, dissolved oxygen
Cloud & DevOps	OpenStack, Docker, Git, GitHub, Linux (Ubuntu, Kali), VMware, Server
Networking & Security	Huawei HCIA-Datacom, Ethical Hacking, Cyber Security Fundamentals
Monitoring & Design	Grafana, Fritzing, SolidWorks
Project Management	Agile, scheduling & budgeting, technical documentation

MAIN PROJECTS & RESEARCH

LoRa Buoy-Based Lake Water Quality Monitoring System — Final Project (2025)

Project Manager & System Analyst · ESP32 · LoRa · Raspberry Pi 4 · MQTT · InfluxDB · Grafana

- Led a three-person team using Agile to deliver a solar-powered buoy prototype that monitors temperature, pH, TDS, and turbidity in real time; managed the work schedule and budget.
- Designed the end-to-end architecture: ESP32 + LoRa on the buoy → LoRa receiver and Raspberry Pi 4 gateway → MQTT → InfluxDB time-series storage → Grafana dashboard running locally, with no internet dependency.
- Defined system requirements, selected sensors (DS18B20, analog pH, TDS, turbidity), and wrote the full technical documentation: requirements, planning, design, implementation, and testing.
- Validated the system in a 5+ hour in-water test: data displayed in Grafana in real time with about 1–2 s latency, plus hourly average, maximum, and minimum aggregation.

Second-Generation Multi-Sensor LoRa-Based IoT Buoy — Final Project (2026)

LoRa · GPS · DO, pH, TDS, Temperature & Turbidity Sensors · Solar Power · Cloud IoT Platform

- Developing the next generation for Lake Toba water quality monitoring: adds a Dissolved Oxygen sensor (five parameters), GPS, and a cloud-based IoT platform with automatic warning notifications.
- Scope driven by first-generation evaluation: sensor calibration with standard solutions, LoRa transmission range testing, and open-water trials.

OpenStack Cloud Platform Installation — Coursework Project (2026)

OpenStack · VMware Workstation · Cloud Computing (IaaS)

- Installed OpenStack on a VMware Workstation virtual machine as a hands-on environment for working with a cloud (IaaS) platform.
- Documented the installation process step by step, with screenshots, in a technical report.

Toba Ethnic Pizza Restaurant Information Website — Final Project

Team of 3 · WordPress · Elementor · MySQL · Apache · WhatsApp Ordering

- Developing a restaurant website that lets customers browse the menu and product information easily.
- Adding an online ordering flow via WhatsApp, available at any time, to widen the restaurant's market reach.

Manalu Village Information System — Community Service (PKM)

WordPress · Elementor · MySQL · Apache · Manalu Village, Pakkat District, Humbang Hasundutan Regency

- Developing a village information system with village profile, village potential, announcements, news, and village services features.
- Helping residents of Manalu Village access village information digitally as part of the community service program.

EDUCATION & ORGANIZATION

D-III Computer Technology — Institut Teknologi Del 2024 – Present

- **Active Member, HIMATEK** (Computer Technology Student Association), IT Del — takes part in work programs focused on developing members' technical skills.
- **IT Del Department of Arts and Culture** — helps organize student creative activities and programs that develop, preserve, and promote arts and culture.

CERTIFICATIONS & AWARDS

- **HCIA-Datacom V1.0** — Huawei ICT Academy
- **HCIA-Storage V5.0** — Huawei ICT Academy
- **Ethical Hacker** — Cisco Networking Academy
- **Introduction to Cyber Security** — Cisco Networking Academy
- **Data Analytics Essentials** — Cisco Networking Academy
- **Aguna Course** — Python, Linux, Network, and Virtual Machine Fundamentals
- **Gemastik** — Pusat Layanan Pembiayaan Pendidikan Kemendikbud Ristek
- **Web Design Competition** — PK-TECHSOFT / HIMA-RPL